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Automated Grounding in FCA by means of LLMs

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Abstract

This paper addresses the grounding problem in Formal Concept Analysis (FCA): suggesting names for anonymous concepts that arise in concept lattices. We propose a methodology that leverages Large Language Models (LLMs) to assign them meaningful names, enriching the lattice while formally preserving its structure, as established by a lattice preservation proposition. An implementation automates the generation of questions for the LLM and the integration of its responses into the formal context. Reliability is assessed through a Knowledge Ablation Test (KAT) based on the semantic similarity between recovered and original attributes. Over seven formal theories comprising 42 attributes, the method recovers 31 within an empirically calibrated tolerance (KAT score 0.73), with a mean cosine similarity above the 0.70 acceptability threshold, showing that the generated labels preserve the semantic content of the originals.

1. Introduction

Extracting meaningful conceptual structures from data is a long-standing challenge in knowledge representation. As datasets grow in scale, automated methods must not only identify patterns but also communicate them in human-understandable terms—a challenge that is particularly acute in formal frameworks, where structural relationships are precisely defined yet their interpretation requires bridging formal representations and natural language.

Formal Concept Analysis (FCA), introduced by Ganter & Wille (1999), is a rigorous framework for discovering conceptual hierarchies within data through lattice structures. Despite its value across software engineering, data mining, and ontology construction, a persistent challenge remains: the grounding problem of assigning meaningful names to concepts that emerge from the analysis but

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lack pre-existing labels. This is especially acute for "anonymous" concepts, defined by unique combinations of attributes without human-understandable designations.

Traditionally this task is addressed manually by domain experts—a process that is time-consuming, subjective, and hard to scale. Recent advances in Large Language Models (LLMs) such as GPT-5, OpenAI (2025), offer new possibilities: trained on vast, diverse corpora, they can understand context, identify semantic relationships, and generalize to novel concept combinations.

Building on our preliminary work, Chan et al. (2023), we proposed an approach that leverages LLMs to automate grounding within ABC,Li et al. (2018). By framing the task as carefully constructed natural-language queries, the method generates appropriate concept labels while preserving the structural integrity of the FCA lattice, offering automation, consistency, and scalability across evolving datasets.

Compared to that preliminary work, this paper makes four contributions. First, we instantiate the grounding methodology within the formal framework of FCA, replacing the domain-specific ABC context with a general formal context (G, M, I) and its concept lattice. Second, we introduce a structured query template that exploits three lattice-level sources—the own attributes of parent concepts, the extension of the anonymous concept (positive examples), and objects in parent extensions not shared by it (counter-examples)—to build precise, automatically generated queries without human intervention. Third, we give a formal argument that the context-update step preserves the lattice structure, so semantic enrichment does not alter the conceptual organisation. Fourth, we introduce the Knowledge Ablation Test (KAT), an instantiation of the knowledge-probing paradigm, Petroni et al. (2019), tailored to FCA, providing a principled, domain-agnostic evaluation grounded on an empirically calibrated similarity threshold.

The paper is organized as follows: Sections 2 and 3 cover background and related work; Section 4 details the methodology and Section 5 its implementation; Section 6 presents a working example; Section 7 defines and applies the validation method; and Section 8 concludes.

2. Background

2.1 Formal Concept Analysis

Formal Concept Analysis (FCA) discovers and visualizes conceptual structures within datasets. Rooted in lattice theory and developed by Ganter & Wille (1999) , it has found applications in software engineering, data mining, and knowledge discovery, Poelmans et al. (2010).

The fundamental components of FCA are based on the initial and core structure:

Definition 1 *A **Formal Context**, K, is a triple (G, M, I) where G is a set of objects, M is a set of attributes, and $I \subseteq G \times M$ is a binary relation, such that $(g, m) \in I$ if the object $g \in G$ holds for the attribute $m \in M$.*

Definition 2 *For a formal context $K = (G, M, I)$, the **derivation operator** $(\cdot)'$ is defined in both directions. For a set of objects $A \subseteq G$:*

$$A' = \{m \in M \mid (g, m) \in I \text{ for all } g \in A\}$$

and for a set of attributes $B \subseteq M$:

$$B' = \{g \in G \mid (g, m) \in I \text{ for all } m \in B\}$$

That is, A' is the set of attributes shared by all objects in A, and B' is the set of objects that share all attributes in B.

From here, we can define:

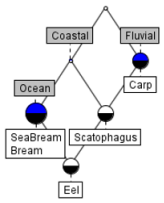
Definition 3 *A **Formal Concept** is a pair (A, B) where $A \subseteq G$ (Extension), $B \subseteq M$ (Intension), satisfying that $A' = B$ and $B' = A$. The set of all concepts is defined to be $\mathcal{T}$.*

For a concept $C = (A, B)$ we write $\text{Ext}(C) := A$ and $\text{Int}(C) := B$. The extension induces a partial order, $C_i \leq C_j$ iff $\text{Ext}(C_i) \subseteq \text{Ext}(C_j)$, giving the set of all concepts $\mathcal{T}$ the structure of a finite lattice. The **Concept Lattice** $\mathcal{L} = (\mathcal{T}, \leq)$ collects all formal concepts of a context ordered by extent inclusion, and can be drawn as a Hasse diagram (see 2.1).

To better illustrate these concepts, we present the following reduced example Ganter & Wille (1999)

A	B	C	D
	Fluvial	Coastal	Ocean
Carp	X		
Scatophagus	X	X	
Bream		X	X
SeaBream		X	X
Eel	X	X	X

(a) Formal Context



(b) Concept Lattice

Figure 1. Formal context (left) and its corresponding concept lattice (right)

We now define further elements needed to address the grounding problem in FCA.

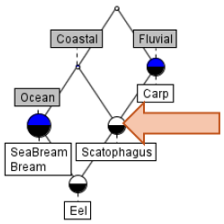
Definition 4 *The **own attributes** of a concept C_i, denoted $A_p(C_i)$, are the attributes that belong to its intension but not to the intension of any strictly larger concept:*

$$A_p(C_i) \;=\; \{m \in \text{Int}(C_i) \mid \forall C_j \in \mathcal{T} :\; C_i < C_j \;\Rightarrow\; m \notin \text{Int}(C_j)\}$$

These attributes distinguish the concept from those above it. Dually, we can define *own objects*. Both decorate the Hasse diagram (fig. 2.1: gray = own attributes, white = own objects).

From here, we can define *anonymous concepts* as those that do not have own attributes in its extensions. More formally:

Definition 5 (Anonymous Concept) *Let C_i be a formal concept in a concept lattice, L. C_i is **anonymous** iff $A_p(C_i) = \emptyset$.*



({*Scatophagus, Eel*}, {*Coastal, Fluvial*})

Figure 2. Anonymous concepts in Concept Lattice

2.2 The Grounding Problem

In our scope, the Grounding Problem refers to the challenge of connecting abstract symbolic representations —concepts, predicates, or rules— to the real-world entities they describe, so that symbols acquire meaning anchored in the external world rather than merely internal to the system. It is particularly relevant when designing ontologies, knowledge graphs, or intelligent agents: without proper grounding, systems risk to be sound but inadequate with respect to the intended domain.

2.3 Grounding FCA

In FCA, grounding is the challenge of assigning meaningful labels to concepts that emerge from the analysis but do not correspond directly to existing attributes. Well-grounded names are essential for the interpretability and practical value of a concept lattice: in line with Explainable AI (XAI) Lipton (2018), clear and semantically coherent naming lets domain experts grasp both the structure of the lattice and the reasoning behind its conceptual organisation, facilitates communication within interdisciplinary teams, and promotes interoperability with other knowledge representation systems. Traditionally this task has been performed manually by domain experts, an approach that faces challenges of scalability, consistency, and subjectivity.

2.4 Large Language Models

Large Language Models (LLMs) are neural models trained on large text corpora to understand, generate, and manipulate human language, with notable examples including the GPT series Radford et al. (2018), BERT Devlin et al. (2018), and their successors.

Beyond fluent language understanding and generation, modern LLMs exhibit strong **task adaptation** and **few-shot learning**, solving novel tasks from only a handful of examples Brown et al. (2020), and they integrate **cross-domain knowledge** acquired during pre-training.

Several of these capabilities make LLMs well suited to the grounding problem in FCA. Their **semantic understanding** lets them interpret the attributes and objects that define a concept; their **contextual processing** situates that concept within the surrounding lattice; and their **knowledge integration** enables them to propose names that are both semantically appropriate and contextually grounded. When guided by well-designed prompts they also name concepts **consistently** and **scale** to large lattices. Our approach leverages these properties to merge the structural rigour of FCA with the semantic depth of LLMs.

3. Related Work

Symbol grounding. The symbol grounding problem was formally characterised by Harnad (1990) as the challenge of connecting abstract symbols to the real-world entities they denote, rather than merely to other symbols. Our work addresses a specific instance within FCA: assigning externally grounded labels to anonymous concepts that emerge from the lattice without any human-assigned designation.

FCA and ontology enrichment. FCA has a well-established role in ontology construction. Stumme & Maedche (2001) pioneered FCA for ontology merging via the FCA-Merge algorithm; Cimiano et al. (2005) learned concept hierarchies directly from text corpora; and Priss (2006) and Poelmans et al. (2010) survey the broad use of FCA in information science and knowledge discovery. None of these addresses the specific problem of assigning meaningful labels to anonymous concepts already present in a lattice—the gap our work fills.

Automated ontology learning. Automated ontology learning from text Maedche & Staab (2001) encompasses concept extraction, taxonomy induction, and relation learning. With large pre-trained language models, Babaei Giglou et al. (2023) propose LLMs4OL, a framework evaluating LLMs on term typing, taxonomy discovery, and relation extraction. Our approach is complementary: whereas LLMs4OL constructs ontologies from scratch, we address a post-hoc labelling problem in which a formal structure already exists but its anonymous nodes lack human-understable names.

LLMs as knowledge stores. Petroni et al. (2019) demonstrated that BERT-class models encode substantial factual knowledge recoverable through cloze-style probing,Taylor (1953), establishing LLMs as implicit knowledge bases, and Pan et al. (2024) provide a roadmap for integrating LLMs with explicit knowledge graphs. In the FCA domain specifically, Chan et al. (2023) were the first to apply LLMs to symbol grounding within the ABC Repair System, Li et al. (2018), framing grounding as a targeted natural-language query task—a direction our work extends by instantiating it within FCA's lattice structure, exploiting its structural relationships (parent attributes, extension objects, and counter-examples) to formulate precise queries, and adding the Knowledge Ablation Test as a principled, domain-agnostic evaluation (Section 7).

4. Methodology

Our approach employs LLMs within a structured, iterative procedure that assigns informative labels to otherwise anonymous concepts in FCA lattices, while preserving the underlying lattice structure and keeping modifications gradual and conceptually coherent.

The process begins by identifying all *anonymous concepts* via a breadth-first traversal from the top of the lattice, so that more abstract concepts are processed before their subconcepts. The relative ordering of siblings does not affect correctness — the Lattice Preservation Proposition guarantees that each update is independent of sibling order — and ties are broken lexicographically by the objects in the extension. Terms are then assigned top-down; after each assignment the context and concept's extensions are recomputed, but the lattice's structure and GET_ANONYMOUS_CONCEPTS

Algorithm 1 Iterative Grounding Process

```
procedure GROUNDING_PROCESS(K,llm)
    L ← Get_Lattice(K)
    Q ← Get_Anonymous_Concepts(L)
    while Q is not empty do
        c ← Q.pop()
        question ← GenerateQuestion(c, L)
        name ← AskName(question, llm)
        K ← Add_Attribute(K, c, name)
        L ← Recompute_Lattice(K)
    end while
end procedure
```

don't need to be refreshed. This progression propagates meaning coherently through the structure, and the resulting assignments are integrated back into the formal context.

To illustrate all phases of the methodology, we will use the formal context shown in Figure 2 and its corresponding lattice.

Identifying Anonymous Concepts

Anonymous concepts are identified with a standard breadth-first traversal of the hierarchy; naming then proceeds iteratively from the most abstract concepts towards more specific ones, ensuring smooth changes to the lattice and consistent naming conventions.

Question Generation and LLM integration

A critical aspect of the methodology is the automatic construction of effective queries. For each anonymous concept, a natural-language question is generated with enough context and constraints to guide the LLM toward an appropriate name and, by design, these questions can be produced automatically by software. The general structure is:

```
What is the term for (SCOPE)
which is (ATR1) and (ATR2)..., as (OBJ1), (OBJ2), (OBJ3),...,
but exclude (CEX1), nor (CEX2), nor ...?
Please, answer with a single term (one word or a short compound
of at most three words).
```

Where:

- SCOPE: Identifies the domain of the formal context (e.g., *fishes*, *animals*). It is read automatically from a `.yaml` configuration file associated with each formal context, requiring no additional user input.
- ATR1, ATR2, etc.: are the Own attributes of the concept's direct parents
- OBJ1, OBJ2, etc.: are Objects in the concept's extension (positive examples) and

- CEX1, CEX2, etc.: are Counter-examples (objects which are in parent extensions, but not in it) to refine the search

In our example, a single anonymous concept remains, for which we generate:

```
what is the term for fishes
   which are Coastal and Fluvial, such as Scatophagus and Eel,
   but exclude Carp and Bream?.
   Please, answer with a single term (one word or a short
   compound of at most three words).
```

Following Chan et al. (2023), we use an LLM (GPT-5 OpenAI (2025)) to answer the question with a single term, which becomes the new attribute name: the question is sent to the LLM API, the response is parsed and validated (a single, unused term), and if necessary the query is reformulated. The answer replied by the LLM (usually) is: **Euryhaline**.

4.1 Context Update and Lattice Preservation

Once a valid term is obtained, the context is updated by adding it as an attribute to M, assigning it to all objects in the extension of the anonymous concept, and recomputing the lattice. Crucially, this preserves the lattice structure: new attributes are introduced only for anonymous concepts and assigned to every object in the concept's extent. The following proposition formalises this invariance.

Proposition 1 (Lattice Preservation) *Let $\mathcal{K} = (G, M, I)$ be a formal context, let $C^* = (A, B)$ be an anonymous concept in $\mathfrak{B}(\mathcal{K})$, and let $m^* \notin M$ be a term generated by the LLM. Define the updated context*

$$\mathcal{K}' \;=\; \big(G,\; M \cup \{m^*\},\; I \cup \{(g, m^*) \mid g \in A\}\big).$$

Then $\mathfrak{B}(\mathcal{K}') \cong \mathfrak{B}(\mathcal{K})$ as a partially ordered set.

Proof 1 *In $\mathcal{K}'$, m^* is possessed by exactly the objects in $A = \mathrm{Ext}(C^*)$. Therefore the extension of every concept in $\mathcal{K}'$ coincides with its extension in $\mathcal{K}$: for a subconcept C_j of C^* (i.e. $\mathrm{Ext}(C_j) \subseteq A$), m^* is added to $\mathrm{Int}(C_j)$ but $\mathrm{Ext}(C_j)$ remains unchanged; for any concept not below C^*, neither intension nor extension changes. Since the partial order on concepts is defined solely by extension inclusion, no order relation is altered, and $\mathfrak{B}(\mathcal{K}') \cong \mathfrak{B}(\mathcal{K})$.*

In our running example, the new attribute is added together with its incidence relation I to the corresponding objects, and the recomputed lattice preserves the original structure while incorporating the new attribute as a proper refinement.

5. Implementation

Our implementation is built entirely in Python, whose ecosystem for knowledge representation and AI allows seamless integration between the two core components of the system: the FCA Engine and the LLM Interface.

5.1 System Architecture

The system comprises two main modules—the **FCA Engine**, which handles FCA operations and lattice manipulation, and the **LLM Interface**, which manages communication with the LLM API—coordinated by a central control script that orchestrates the overall process and data flow.

Main Control Script

The main control script integrates both modules into a coherent pipeline: it initialises the FCA Engine and the LLM Interface, executes the iterative naming algorithm, supervises data exchange between the modules, and manages logging and error reporting.

FCA Engine

The FCA Engine handles all formal-context manipulation and concept-lattice management. It is implemented with the `concepts` library, Bank (2023), which provides native Python support for loading formal contexts, generating concept lattices, and identifying own objects. It first loads the context, computes the lattice, and extracts the relevant elements of each anonymous concept—parent attributes, extension objects, and counter-examples—for the question-generation module. Once the LLM Interface returns a valid term, it updates the context by adding the new attribute and its incidence relation, and recomputes the lattice to verify that its structure is preserved.

LLM Interface

The LLM Interface handles all communication with the LLM through its API, transmitting the queries built by the FCA Engine as plain text. Upon receiving a response, it validates that the output is a valid term of no more than three words: while the prompt requests a single word, compound expressions of up to three words are accepted, and longer outputs are treated as explanatory descriptions and rejected, in which case the query is automatically reformulated.

All experiments were conducted using GPT-5 (OpenAI) with default configuration parameters. In the GPT-5 model family, `temperature` is fixed at 1 and is not user-configurable; `top_p` sampling is disabled by default; and `frequency_penalty` and `presence_penalty` are both set to 0. Crucially, `reasoning_effort` was set to `none`, disabling the extended chain-of-thought reasoning mechanism introduced in this model family. No additional system prompt or role instruction was supplied beyond the query template described in Section 4.

Error Handling and Logging

The system logs each step of the naming process, applies recovery strategies for API failures or invalid responses, and periodically stores intermediate results so that the process can be resumed after an interruption.

6. Working example: Animals Classification

To illustrate our approach, we apply the methodology to a case study in animal classification, Ganter & Wille (1999), showing how it enriches an FCA lattice with meaningful concept names. Like most real-world formal contexts, this one is informationally sparse: each animal is reduced to a small set of binary attributes with no textual description. This austerity, representative of real-world conditions, places significant demands on the grounding process and motivates the question template of Section 4, which compensates for the lack of explicit semantic information by exploiting the structural relationships of the lattice.

6.1 Initial Context

We begin with a formal context describing a variety of animals (Fig. 3)—from mammals and birds to aquatic creatures and reptiles—with attributes covering their physical traits, behaviour, habitat, and relationship with humans.

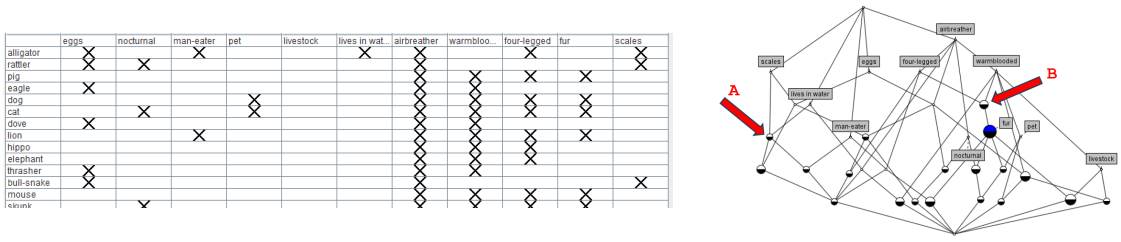


Figure 3. Animals context

From this context we generate the initial concept lattice, focusing for brevity on the subset of concepts subject to our naming process.

Identifying Anonymous Concepts

Several anonymous concepts in the initial lattice would benefit from meaningful names. We present two representative cases—concept A, ({*Lives in water, Scales*}, {*Barracuda,...*}), and concept B, ({*AirBreather, Four-Legged, Warm-Blooded*}, {*Elephant, Hippo,...*})—illustrating the key steps: instantiating the question template, integrating the LLM response, and updating the lattice before the next iteration. The exhaustive application to the full corpus and its quantitative evaluation are reported in Section 7.

6.2 Iterative Naming Process

At each step, assigning a name may alter the attributes characterising a concept and its subconcepts, so the context is recomputed to keep subsequent queries consistent with the updated lattice. This naming–updating cycle reflects the top–down strategy of our methodology, where names assigned to higher-level concepts shape the questions for more specific ones, so that each new attribute simplifies later queries and improves their precision.

6.2.1 Naming Concept A

For Concept A ({*Lives in water, Scales*}, {*Barracuda,...*}), we generate the following question for the LLM:

```
what is the term for animals which are Aquatic and have scales,
as Barracuda, but not bull-snake, nor rattler, nor dolphin,....
Please, answer with a single term (one word or a short compound
of at most three words).
```

LLM Response: “Fish"

We add “Fish” as a new attribute, associating it with Barracuda, Stingray, Whale, etc.; validation of this and all assignments is left to experts, interactively or at the end. Since this concept sits higher in the lattice than other anonymous ones, its new label is reused later when naming its subconcepts.

6.2.2 Naming Concept B

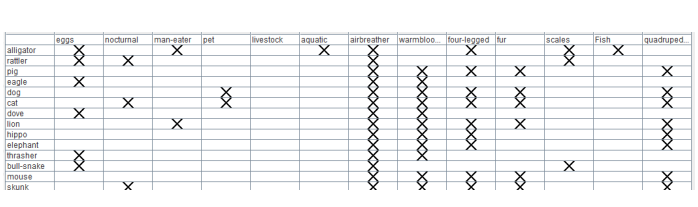
Concept B has intension ({*AirBreather, Four-Legged, Warm-Blooded*}, {*Elephant, Hippo,...*}), where *AirBreather* is inherited from a parent and is not an own attribute of B. Its own attributes, *Four-Legged* and *Warm-Blooded*, are the ones used to characterise B in the question:

```
what is the term for animals which are Four-Legged and
Warm-Blooded, as Elephant and Hippo, but exclude Lizard,
Alligator.
Please, answer with a single term (one word or a short compound
of at most three words).
```

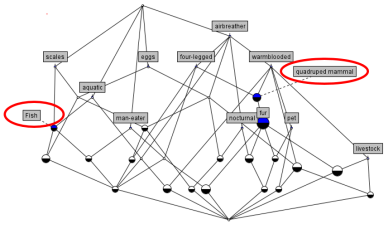
The LLM answers “Quadruped Mammals”, a two-word compound within the three-word limit, clearly corresponding to the animals in the concept’s extension.

6.3 Context and Lattice updated

After applying this naming process, we obtain an enriched formal context (fig. 4a):



(a) Updated context



(b) Updated Concept Lattice

Figure 4. Context and Lattice updated

The updated lattice (Fig. 4b) now includes the new attributes, providing meaningful, domain-specific names aligned with existing animal classifications.

7. Validation: Knowledge Ablation Test

Validating the outputs of Large Language Models (LLMs) is a central concern, given their well-documented tendency to generate subtle errors, flawed reasoning chains, and unsupported claims, Ji et al. (2023)Huang et al. (2023). Existing strategies range from information-retrieval–based methods Lewis et al. (2020) Baek et al. (2023) and symbolic verifiers Qu et al. (2024) to auxiliary models acting as evaluators Kadavath et al. (2022). Particularly informative are methods that introduce controlled perturbations to the knowledge base: closely related to probing and leave-one-out analyses Petroni et al. (2019), the *knowledge ablation* paradigm deliberately removes an informational unit from the knowledge set and evaluates the model's ability to reconstruct it from the remaining context, thereby distinguishing memorised from genuinely reasoned knowledge. Building on this principle, we introduce the *Knowledge Ablation Test* (KAT) as a systematic procedure for validating our grounding methodology.

The core hypothesis is straightforward: if the methodology produces semantically appropriate labels for anonymous concepts, it should also recover — or closely approximate — attribute names that were originally present in the formal context but deliberately removed. The procedure has three stages: we calibrate a semantic acceptability threshold from synonym distributions; we systematically ablate attributes to produce incomplete contexts; and we apply the grounding procedure to those contexts, comparing the generated labels against the original attributes through a semantic similarity function, Reimers & Gurevych (2019).

7.1 Formalisation

The KAT evaluates the ability of an LLM to reconstruct missing information from a partial knowledge set. Let $A = \{a_1, a_2, \ldots, a_n\}$ be a finite set of knowledge units; in the FCA setting that concerns us, A corresponds to the set of attributes M of a formal context. Given a language model LM and a knowledge base A, we define a knowledge-dependent task as

$$T : LM \times A \longrightarrow R,$$

where R is the space of admissible completions of A proposed by LM.

The KAT comprises three phases:

- **Ablation.** For any element $a_i \in A$, the reduced knowledge base $A_{-i} = A \backslash \{a_i\}$ is constructed by removing a_i from the input.
- **Recovery.** The model is prompted to recover $\hat{a}_i$ from the incomplete context A_{-i}:

$$T(LM, A_{-i}) \longrightarrow \hat{a}_i.$$

- **Evaluation.** The recovered $\hat{a}_i$ is compared with the original a_i via a semantic similarity measure $\mathrm{sim}(\cdot, \cdot)$ and threshold θ:

$$\mathrm{Eval}(a_i, \hat{a}_i) = \begin{cases} 1 & \text{if } \mathrm{sim}(\hat{a}_i, a_i) \geq \theta, \\ 0 & \text{otherwise.} \end{cases}$$

Aggregating over the full knowledge base, the overall score of the test is defined as

$$\mathrm{KAT}(LM, A) \;=\; \frac{1}{|A|}\sum_{i=1}^{n} \mathrm{Eval}\big(a_i,\, T(LM, A_{-i})\big), \tag{1}$$

which quantifies the model's capacity to reconstruct missing knowledge—here, the suitability of the grounding method.

7.2 Similarity Function and Tolerance Margin

Two design choices govern the KAT: the similarity measure $\mathrm{sim}(\cdot,\cdot)$ and the threshold θ. Since the terms returned by the LLM rarely coincide lexically with the original attribute—synonyms and paraphrases are common—a purely lexical comparison would strongly bias the evaluation against the method.

Semantic similarity. We therefore assess similarity at the semantic level. Each candidate term is mapped to a dense representation through a pre-trained sentence-embedding model, and similarity is measured as the cosine between embeddings. We use the `all-MiniLM-L6-v2` model from SENTENCETRANSFORMERS, Reimers & Gurevych (2019), which offers a favourable quality/cost trade-off. Given two terms x and y with embeddings $\mathbf{v}_x, \mathbf{v}_y$,

$$\mathrm{sim}(x, y) \;=\; \frac{\mathbf{v}_x \cdot \mathbf{v}_y}{\|\mathbf{v}_x\|\,\|\mathbf{v}_y\|}, \qquad d(x, y) \;=\; 1 - \mathrm{sim}(x, y), \tag{2}$$

so that values close to 1 indicate strong semantic proximity.

Calibrating the threshold. The threshold θ must reflect the typical semantic distance between a term and its admissible alternatives in the target domain. Rather than fixing it a priori, we calibrate it experimentally, using an LLM—rather than a static thesaurus—so that the synonyms are contextually appropriate for each formal context.

Experimental corpus. The corpus was drawn from two public repositories: the FCA Repository (`https://fcarepository.org/`) and the collection maintained by Uta Priss (`https://upriss.github.io/fca/examples.html`) Priss (2006). Both publish the contexts together with the `.yaml` description files that supply the SCOPE metadata required by our methodology. We applied a two-stage filter: we discarded non-binary contexts (many-valued or fuzzy), since our methodology assumes a binary incidence relation, and synthetic contexts whose objects and attributes lack human-readable semantics. After filtering, seven contexts remained, comprising 42 attributes.

For each attribute $a_i \in M$ we used an LLM-based synonym generator to produce one contextually appropriate synonym $\mathrm{Syn}(a_i) = \hat{a}_i$; the resulting 42 (attribute, synonym) pairs form a calibration set reflecting the natural terminological variability of the corpus. For each pair we computed the cosine similarity between embeddings. The resulting distribution (Fig. 5) concentrates above 0.7, with most pairs between 0.75 and 0.95.

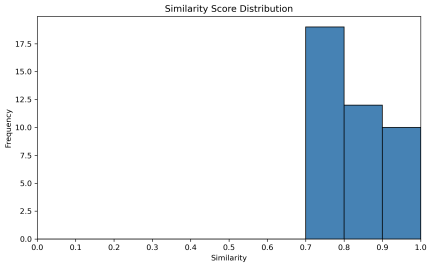


Figure 5. Distribution of cosine similarity scores between original attributes and their LLM-generated synonyms (42 terms drawn from 7 formal contexts).

We therefore adopt $\theta = 0.7$ as the operational threshold: a recovered term is semantically acceptable whenever its cosine similarity to the original attribute is at least 0.7. This anchors the acceptability criterion on the empirical distribution of synonym similarities rather than on an arbitrary cut-off.

7.3 Application to Formal Concept Analysis

We now instantiate the general KAT formulation for FCA. Given a formal context $\mathcal{K} = (G, M, I)$ and its concept lattice $\mathfrak{B}(\mathcal{K})$, the knowledge units to be ablated are precisely the attributes in M. For each $a_i \in M$ we replace the attribute by an anonymous one and construct the *ablated context*

$$\mathcal{K}_{-i} \;=\; \big(\, G,\; M \setminus \{a_i\},\; I \cap (G \times (M \setminus \{a_i\}))\,\big), \tag{3}$$

and derive the lattice $\mathfrak{B}(\mathcal{K}_{-i})$. Removing an attribute may produce one or several anonymous concepts; among them we identify the concept whose extension matches the objects that originally possessed a_i,

$$c^* \;=\; \big(\,\{g \in G \mid (g, a_i) \in I\},\, \cdot\,\big) \;\in\; \mathfrak{B}(\mathcal{K}_{-i}), \tag{4}$$

and apply the grounding methodology of Section 4 to c^*. Let $\hat{a}_i$ denote the term proposed by the LLM. Recovery is declared *successful* when

$$\mathrm{sim}(a_i, \hat{a}_i) \;\geq\; \theta, \tag{5}$$

and a *miss* otherwise. We retain the calibrated threshold $\theta = 0.7$, and repeat the procedure for every non-trivial attribute of each formal context, producing the global score of Eq. (1).

7.4 Experimental Results

The ablation–recovery–evaluation cycle was executed for every attribute of the formal contexts in our case studies, recording each recovered term and its cosine similarity to the original. The resulting distribution is shown in Fig. 6.

The recovered terms consistently fall within — or close to — the tolerance margin: in most cases their similarity to the original attribute is comparable to, or higher than, the mean attribute–synonym similarity, even though many are not exact lexical matches. This indicates that the grounding process produces semantically valid labels aligned with the natural terminological variability of the domain.

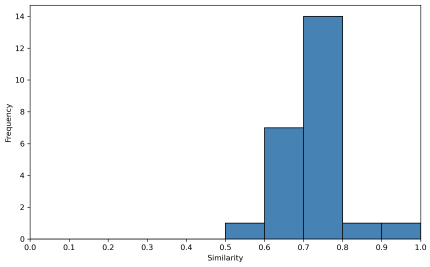


Figure 6. Distribution of cosine similarities between ablated attributes and the terms recovered by the grounding method.

Applying Equation 1 to the complete corpus of 42 attributes yields the following aggregate result:

$$\mathrm{KAT}(LM, A) \;=\; 0.73 \tag{6}$$

That is, 31 out of 42 attributes were successfully recovered within the tolerance margin ($\theta = 0.7$), with a mean cosine similarity of $\mu = 0.82$ ($\sigma = 0.07$), well above the threshold. Since this mean substantially exceeds both the threshold and the mean synonym similarity from calibration, the method does not merely match a synonym generator but consistently produces terms semantically close to the originals.

Absolute vs. relative evaluation. The KAT is an *absolute* validation instrument, not a comparative benchmark: its goal is to establish whether the method genuinely recovers semantic information present in the original context. The natural implicit baseline is *uninformed recovery* — a term drawn at random from the domain vocabulary — which under the all-MiniLM-L6-v2 embedding space yields cosine similarities clustering around 0.3–0.4, well below both the threshold $\theta = 0.7$ and the mean recovery similarity $\mu = 0.82$. This substantial gap confirms that the methodology captures structured conceptual information rather than producing arbitrary outputs.

Boundary and failure cases. Boundary cases — those with $\mathrm{sim}(a_i, \hat{a}_i)$ close to θ — typically correspond to valid but less conventional labels that a domain expert would still accept. The few failures can be attributed to three patterns: (a) highly specific or technical attributes for which the LLM's pre-training corpus offers limited coverage; (b) insufficient discriminating context after ablation, where the parents of c^* and its extension do not jointly constrain the target tightly enough; and (c) hypernym/hyponym substitution, where the LLM proposes a term broader or narrower than the original.

The alignment between recovered terms and the empirically calibrated tolerance margin is strong evidence that the grounding methodology preserves much of the semantic information contained in the original context, even under the incomplete-information conditions of knowledge ablation. Finally, the KAT framework is itself an additional contribution, providing a general methodology for evaluating semantic-reconstruction capabilities in LLM-based symbolic grounding systems.

8. Conclusions and future work

This work has introduced a methodology that integrates Large Language Models into the grounding process of Formal Concept Analysis, automatically assigning meaningful labels to anonymous concepts while preserving the structural integrity of the concept lattice, and establishing a foundation for further integration of symbolic and language-based AI in knowledge representation.

The approach offers several advantages: the LLM-generated terms enhance the interpretability of the lattice with domain-relevant labels that align with expert terminology; the Lattice Preservation Proposition guarantees that adding these attributes leaves the underlying structure unchanged; and the method draws on the broad knowledge embedded in contemporary LLMs, making it applicable across diverse domains.

Several limitations must be acknowledged. The quality of the generated labels depends on the domain coverage and reliability of the LLM, which varies across specialised or highly technical contexts; the statistical nature of LLM responses can introduce uncertainty, occasionally requiring reformulations; the binary attribute representation intrinsic to FCA imposes simplifications that may obscure nuanced domain characteristics; and, relatedly, the method inherits the classical view of concepts as sets of necessary and sufficient attributes, so we make no claim of psychological validity: the labelling task is one of lexicalisation rather than concept formation, and the resulting terms reflect aggregate naming conventions instead of modelling human categorisation, which is known to be graded and typicality-based (Rosch (1975)).

Potential biases in the training data likewise remain a concern. Promising directions for future work include interactive tools for managing and validating LLM-generated terms, the use of LLMs within attribute-exploration frameworks, improvements in prompt engineering and query formulation, systematic comparisons across different LLM architectures and sizes, and human-subject evaluation contrasting model-generated labels with terms elicited from domain experts.

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